

For **Question 5**, we wanted to understand whether company size affects salary outcomes.

From the data, we see a very clear and consistent trend — as company size increases, salary also increases. Startups have the lowest average salary at around \$127K, followed by small, medium, and large companies, and finally enterprise companies at the top with about \$169K.

So overall, there is a difference of more than \$42K between startups and enterprise companies.

What's important here is that this relationship is monotonic, meaning every step up in company size leads to an increase in salary. There are no drops or inconsistencies in between.

This suggests that larger companies likely have more resources, structured compensation systems, and higher budgets for salaries compared to startups.

So the key takeaway is that company size is a strong and consistent factor influencing salary, and choosing larger organizations can significantly increase earning potential.

For **Question 6**, we explored whether skills and certifications translate into higher pay.

We found that both skills and certifications do have a positive impact on salary, but the effect is relatively small.

For example, increasing the number of skills from 1 to 19 only increases salary by about \$15K. Similarly, certifications increase salary by around \$8K from lowest to highest.

When we look at the correlation values, they are quite low — 0.13 for skills and 0.07 for certifications — which means these are weak predictors of salary.

So even though they help, they are not major drivers compared to factors like experience or location.

We also analyzed whether **industry sector affects salary**, and the result was surprising.

We found that the difference between the highest and lowest average salaries across all industries is only about **\$594**.

This is extremely small compared to other factors we studied.

So what this means is that, within tech roles, the industry you work in does not significantly affect your salary.

To put it in perspective, we saw differences of over \$80K based on location and over \$50K based on job title, but for industry, it's less than \$600.

So the key takeaway is that **industry is almost irrelevant as a salary predictor in this dataset**, especially when compared to stronger factors like experience, location, and job role.

For **Question 7**, we examined whether education pays off more for senior workers compared to junior workers.

We divided the data into two groups — junior workers with 0 to 7 years of experience, and senior workers with 8 to 20 years of experience.

For junior workers, we found that salary differences across education levels are relatively small. This means that early in a career, education does not create a large salary gap.

However, for senior workers, the difference becomes much more significant. The gap between lower education levels like diploma and higher levels like PhD increases noticeably.

This shows that education becomes more valuable as experience increases. In other words, higher education combined with more years of experience leads to the highest salaries.

So the key takeaway here is that education alone is not a strong factor early on, but over time, when combined with experience, it plays a much bigger role in salary growth.